Summary in Graph

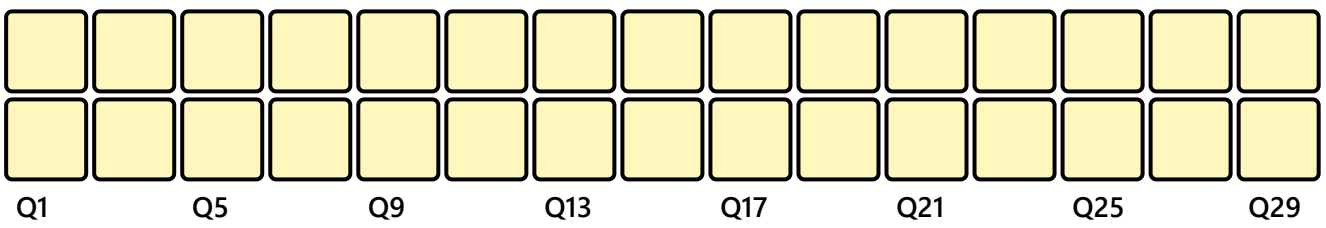
Exam Summary (GO Classes CS Test Series 2026 | CO and Architecture | Subject Wise Test 2)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (15 + 15)
Total Marks:	45 (15 + 30)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (37) out of 370 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

A characteristic of a RISC architecture is:

A. A high-risk architecture as the hardware market it evolves very quickly
B. It has a large set of complex instructions
C. The architecture consists of thousands of processors

D. It has a small set of simple instructions

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

By segmenting a processor, transforming it into a pipeline, we obtain:

A. reduction in the number of cycles required to execute an instruction
B. reduction in the number of cycles required to execute a program
C. reduction in the number of cycles required to handle an exception
D. reduction in the number of cycles required to handle an interrupt

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | CO & Architecture

For floating-point number representation in the IEEE standard, which of the following statements are true?

I. When the fraction and the exponent are zero, the represented number is zero.

II. When the exponent is zero and the fraction is different from zero, the represented number is denormalized.

III. When all exponent bits are equal to one and the fraction is zero, the number is $+\infty$ or $-\infty$.

IV. When all exponent bits are equal to one and the fraction is different from zero, the representation is not a number (NaN).

A. I

B. II

C. III

D. IV

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

A processor has the following memory hierarchy: a cache with access latency of $1\ \mathrm{ns}$ and a main memory with access latency of $100\ \mathrm{ns}$. Access to main memory is performed only when the value is not found in the cache.

The MAXIMUM acceptable cache miss rate so that the average memory access time is less than or equal to $2\ \mathrm{ns}$ is:

A. 10%

B. 5%

C. 50%

D. 1%

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

The use of cache memories is very important for the performance of modern processors.

Analyze the statements below regarding the use of cache memories.

I. In a direct-mapped cache, a memory block can be placed in any position (entry) of this cache memory.
II. In the write-back policy, the modified block is updated in main memory only when it is replaced.
III. Using associativity in cache memories serves to reduce the number of conflict misses.

From the analysis, it can be concluded that:

A. the three statements are false.
B. the three statements are true.
C. only statement I is true.
D. only statements II and III are true.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Numerical Type | Award: 1 | Penalty: 0 | CO & Architecture

A processor has five pipeline stages. Suppose that each stage of the processor (fetch, decode, execute, reading or writing data in memory, and writing to the register file) is executed in $5\ \mathbf{ns}$. The total time(in ns) for 5 instructions to be executed in the pipeline, assuming there are no data dependencies between the instructions, is _____ .

Your Answer: | Correct Answer: 45 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

Computer memory is organized in levels. Mark the option that correctly presents the structures found at the highest level of this hierarchy.

A. $\mathrm{L1}$ cache.

B. $\mathrm{L2}$ cache.

C. Hard disk.

D. Processor registers.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

About pipelines, mark the correct option.

A. Each pipeline stage has its own execution time.

B. A pipeline needs registers to store data between stages.

C. Data dependencies will always stall the pipeline.

D. The pipeline is stalled when executing a branch instruction.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

The MESI protocol (also known as the Illinois protocol) is a widely used cache coherence and memory coherence protocol. What are the four cache-line states according to the MESI protocol and what are their respective meanings?

A. **Expanded**: the cache line was modified (it is double the main memory) and is present in every cache.
Dedicated: the cache line is destined for main memory and is not present in any other cache.
Replicated: the cache line is replicated in main memory and may be present in another cache.
Finite: the cache line contains valid data only in main memory.

B. **Modified**: the cache line was modified (different from main memory) and is present only in that cache.
Exclusive: the cache line is the same as in main memory and is not present in any other cache.
Shared: the cache line is the same as in main memory and may be present in another cache.
Invalid: the cache line does not contain valid data.

C. **Expanded**: the cache line was expanded (larger than main memory) and is present in every cache.
Exclusive: the cache line is the same as in main memory and is not present in any other cache.
Replicated: the cache line is replicated in main memory and may be present in another cache.
Finite: the cache line contains valid data only in main memory and the cache is limited.

D. **Modified**: the cache line was duplicated (different from main memory) and is present in every cache.
Dedicated: the cache line is destined for main memory and is not present in any other cache.
Shared: the cache line is different from main memory and may be present in another cache.
Finite: the cache line contains valid data only in main memory.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

Consider a processor that has 16-bit byte addressing. Its cache is direct-mapped, with a capacity of 128 words and blocks of 8 words. The address is divided into fields to access the cache according to the following representation:

- Tag: i
- Index: ii
- Word within the block: iii
- Byte within the word: iv

The number of bits in each of the fields above, that is, the values of $\mathrm{i}, \mathrm{ii}, \mathrm{iii}$, and iv, respectively, are:

A. $12, 6, 5, 3$

B. $8, 4, 3, 1$

C. $8, 6, 5, 3$

D. $12, 4, 3, 1$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

In a 32-bit machine, we split the virtual address into 4 segments as follows:

A 3-level page table is used, so that the first 10 bits are for the first level, and so on. Therefore, what is the size of a page table for a process that has $\mathbf{256}$ KB of memory starting at address 0 ?

A. 6480 bytes
B. 4608 bytes
C. 3240 bytes
D. 2408 bytes

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

A system bus normally consists of around 50 to hundreds of separate lines. Each line has a particular meaning or function. Although there are many different bus designs, in any bus the lines can be classified into functional groups. Mark the INCORRECT alternative about the bus.

A. The data lines provide a path for moving data between the system modules. These lines, collectively, are called the data bus.
B. The memory lines serve to reference data that are stored in memory and make a direct reference to the bus that is in use. Used for accessing data from disk to memory.
C. The address lines are used to designate the source or destination of data on the data bus. In addition, the address lines are generally also used to address I/O ports.
D. The control lines are used to control access to and the use of the data and address lines. Since the data and address lines are shared by all components, a means of controlling their use is required.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

A cache has used a word from a memory block ranged between $0 - 63$. If the same word is required soon then it will exploit

A. Spatial locality of reference

B. Temporal locality of reference

C. Capacity miss

D. None of the above

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

Which of following may not definitely occur in an instruction cycle?
(i) Fetch cycle
(ii) Execution cycle
(iii) Indirect cycle

A. (i) only
B. (i) and (ii)
C. (iii) only
D. (ii) and (iii)

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

A certain machine uses expanding opcode. It has 16-bit instructions and 6-bit addresses. It supports one address and two address instructions only. If there are 'n' two address instructions, the maximum number of one address instruction is

A. $2^\wedge 16 - n$

B. $2^\wedge 10 - n$

C. $\left(2^\wedge 4 - n\right) \times 2^\wedge 6$

D. $2^\wedge 10$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Consider the following statements:

I. A modification to a CPU caused the cycle time and CPI to increase by 10%, while the number of instructions executed for a given application decreased by 20%. We can conclude that the execution time of this application will remain unchanged.
II. A page fault occurs when the entry corresponding to the requested page is not found in the translation lookaside buffer.
III. To store the same amount of data, a direct mapped cache is typically smaller than a set associative cache, assuming blocks of the same size.
IV. Increasing the block size of a cache increases the advantages obtained with spatial locality.
V. Virtual memory typically uses the write-through strategy instead of the write-back strategy.

Which alternatives are true?

A. Only statements I, II, III, and IV are true.

B. Only statements I, III, and IV are true.
C. Only statements II, III, and IV are true.
D. Only statements III and IV are true.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Consider two systems $\mathbf{A}$ and $\mathbf{B}$ composed of a processor, cache, and memory whose only difference is the data cache. The data caches have in common 2-byte words, capacity (for example, 2 KBytes), block size (for example, 8 bytes per line), and are implemented with the same technology, but with different organizations as defined below:
(Cache of A) Direct-mapped cache, using write-through and no-write allocate policies (writes do not use the cache)
(Cache of B) 4-way set-associative cache, using write-back, write-allocate, and LRU policies

Consider the following statements for systems A and B executing the same typical program:

I. System A must have a higher miss rate than B
II. System B must have a lower hit rate than A
III. The data cache of A is faster than that of B
IV. The data cache of A is simpler to implement than that of B
V. On average, a data write in system A is faster than in B
VI. The data caches of A and B have the same number of lines

Which statements are true?

A. Only statements (II), (III) and (IV) are true
B. Only statements (I), (III),(IV) and (VI) are true
C. Only statements (I), (III) and (IV) are true
D. Only statements (II), (V) and (VI) are true

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

A **branch history table (PHT)** is a small memory placed in the instruction fetch stage, indexed by the most significant bits of the address of the instruction being fetched. Each entry in the PHT encodes the prediction of the outcome of the next execution of any branch instruction that indexes that table entry, based on the outcome of a **$\mathbf{2}$-bit saturating counter.**

Assuming $\mathbf{00}$ as the initial value of the counters, **MARK** the prediction of each branch $(\mathrm{A}, \mathrm{B},$ and $\mathrm{C})$ after checking the following outcome sequences for the same branch instruction: $(\mathbf{T} =$ taken, $\mathbf{N} =$ not taken$)$.

$\mathrm{A}: \mathrm{N\ T\ T\ T\ N\ T\ T\ T\ N\ N\ T\ T\ N\ N\ T\ T\ N}$

$\mathrm{B}: \mathrm{N\ T\ N\ N\ N\ N\ N\ N\ N\ N\ T\ N\ T\ T\ N\ N\ T}$

$\mathrm{C}: \mathrm{T\ N\ T\ T\ T\ T\ N\ N\ N\ T\ T\ N\ N\ N\ T\ T\ T}$

A. Taken, Not Taken, Taken

B. Taken, Taken, Taken

C. Not Taken, Not Taken, Not Taken

D. Not Taken, Taken, Not Taken

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Consider the set of operations described below, in register-transfer language.
$\mathrm{F1: REM \leftarrow CP;}$
$\mathrm{F2: RDM \leftarrow M[REM], CP \leftarrow CP + 1;}$
$\mathrm{F3: RI \leftarrow RDM;}$
Data: REM is the memory address register; RDM is the memory data register; RI is the instruction register; CP is the program counter; and $\mathrm{M[X]}$ is the memory contents at address X.

Mark the option that indicates which phase of instruction processing in a pipelined machine corresponds to the described set of operations.

A. Instruction decoding.
B. Operand fetch.
C. Instruction execution.
D. Instruction fetch.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

A processor has an operation cycle equal to $\mathbf{20\ ns}$. It spends two cycles to fetch an instruction, one cycle to decode each instruction, two cycles to fetch the required operands, and three cycles to execute the instruction and store the corresponding result in some register.

If the organization of this machine is strictly sequential, what will be the execution time of one instruction?

If we use $\mathbf{2\ ns}$ delay latches and consider each module as indivisible, what will be the pipeline period if the machine is organized as a four-stage structure?

A. Respectively $20\ \mathrm{ns}$ and $22\ \mathrm{ns}$.

B. Respectively $20\ \mathrm{ns}$ and $42\ \mathrm{ns}$.

C. Respectively $20\ \mathrm{ns}$ and $62\ \mathrm{ns}$.

D. Respectively $160\ \mathrm{ns}$ and $62\ \mathrm{ns}$.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Match the left column with the right column

(I) Multicore	(A) Multiple pipelines operating in parallel.
(II) Superpipeline	(B) Out-of-order execution of instructions in a pipeline.
(III) Superscalar	(C) Pipelines with a large number of stages.
(IV) Dynamic pipeline	(D) Multiple processors sharing an address space.
(V) Multiprocessors	(E) Multiple processors in a single package.

Choose the option that contains the correct matching.

A. $\mathrm{I} - \mathrm{B},\ \mathrm{II} - \mathrm{A},\ \mathrm{III} - \mathrm{C},\ \mathrm{IV} - \mathrm{E},\ \mathrm{V} - \mathrm{D}.$

B. $\mathrm{I} - \mathrm{C},\ \mathrm{II} - \mathrm{A},\ \mathrm{III} - \mathrm{B},\ \mathrm{IV} - \mathrm{D},\ \mathrm{V} - \mathrm{E}.$

C. $\mathrm{I} - \mathrm{D},\ \mathrm{II} - \mathrm{E},\ \mathrm{III} - \mathrm{B},\ \mathrm{IV} - \mathrm{A},\ \mathrm{V} - \mathrm{C}.$

D. $\mathrm{I} - \mathrm{E},\ \mathrm{II} - \mathrm{C},\ \mathrm{III} - \mathrm{A},\ \mathrm{IV} - \mathrm{B},\ \mathrm{V} - \mathrm{D}.$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

A machine $\mathrm{M1}$ operates at $1400\ \mathrm{MHz}$ and has 3 types of instructions: A, B, and C, which take $1, 2$, and 4 cycles, respectively. A certain program P executed on this machine used 20% type A instructions, 30% type B instructions, and 50% type C instructions. A machine $\mathrm{M2}$ also has 3 types of instructions: D, E, and F, which take $3, 4$, and 5 cycles, respectively. Program P, when executed on $\mathrm{M2}$, used 30% type D instructions, 40% type E instructions, and 30% type F instructions.

Mark the option that correctly gives the operating frequency that machine $\mathrm{M2}$ must have so that program P runs in the same time on both machines.

A. $1.6\ \mathrm{GHz}$

B. $1.8\ \mathrm{GHz}$

C. $2.0\ \mathrm{GHz}$

D. $2.2\ \mathrm{GHz}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

A computer with a Reduced Instruction Set Computer (RISC) is a processor architecture line that favors a simple and small set of instructions that take approximately the same amount of time to execute. Typical characteristics of a RISC organization are considered to be:

A. offering support for high-level languages and facilitating compiler development.

B. providing the computer with a complex set of instructions and improving program execution.

C. having few registers and specialized registers.

D. optimizing the instruction pipeline and presenting a limited instruction set with fixed format.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 05sec | Discuss

Q #24 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Consider a computer in which the $level\ 0$ interpreter (hardware interpreting microcode) takes 20 nanoseconds $(20 \times 10^{-9}$ seconds) per microinstruction, and the $level\ 1$ interpreter (microcode interpreting the conventional machine language) requires 25 microinstructions to interpret one conventional machine instruction. A test program, already written in conventional machine language, takes 50 seconds to execute.

a. How long will the test program take to execute if the time to execute one microinstruction is increased to 30 nanoseconds?

b. How long will the test program take to execute if a conventional machine instruction needs only 15 microinstructions to be interpreted?

A. a) $\mathrm{T} = 60$ seconds and b) 15 seconds.

B. a) $\mathrm{T} = 65$ seconds and b) 20 seconds.

C. a) $\mathrm{T} = 70$ seconds and b) 25 seconds.

D. a) $\mathrm{T} = 75$ seconds and b) 30 seconds.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Given a disk rotating at $\mathbf{2,400}$ RPM, with $\mathbf{120}$ cylinders and $\mathbf{6}$ tracks per cylinder, each track has $\mathbf{16}$ sectors and each sector has $\mathbf{512}$ bytes. Considering that the average seek time is $\mathbf{60\ ms}$, the seek time between adjacent cylinders is $\mathbf{10\ ms}$, and the transfer time is $\mathbf{15\ ms}$, how much time will be necessary to read $\mathbf{10}$ sectors?

A. 0.0485 seconds.
B. 0.0685 seconds.
C. 0.0885 seconds.
D. 0.1285 seconds.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Consider a cache memory with a cache line size of $\mathbf{16}$ bytes and a main memory that requires $\mathbf{20\ ns}$ to transfer an 8-byte word to any cache line that is written at least once before being evicted from the cache. What is the average number of times the cache line must be written before it is evicted for a write-back cache to be more efficient than a write-through cache?

A. 4 times before being replaced, then write-back is more efficient.
B. 8 times before being replaced, then write-back is more efficient.
C. 12 times before being replaced, then write-back is more efficient.
D. 16 times before being replaced, then write-back is more efficient.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

A single precision floating point number contains the sequence of bits $10011111100000000000000100000000000000000$. Information is stored in the following left to right order: sign bit, exponent (bias 127) and mantissa.

Which of the following representation in decimal is equivalent?

A. $2^{31}\left(1+2^{-12}\right)$

B. $(-1) * 2^{31}\left(1+2^{-12}\right)$

C. $(-1) * 2^{-65}\left(1+2^{-10}\right)$

D. $(-1) * 2^{96}\left(1+2^{-11}\right)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

A computer has a $\mathbf{256K}$ bytes, $\mathbf{8}$ way set associative, write back data cache with block size of $\mathbf{32}$ bytes. The processor sends $\mathbf{40}$ bit address to the cache controller. Each cache tag directory entry contains, in addition to address tag, $\mathbf{4}$ valid bits, $\mathbf{1}$ modified bit, and $\mathbf{2}$ replacement bits. What is the size of the cache directory?

A. 64 K bits

B. 256 K bits

C. 32 K bits

D. 68 K bits

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Consider the 4 stage pipeline with the following stages: S1: Instruction fetch. S2: Instruction decode. S3: Execute. S4: Write back. Program consists of 16 instructions (I1, I2, ... I16), in which I6 is an unconditional branch instruction that transfers control to I12. In the pipeline, the branch target address will be available at the end of the execute stage. Each instruction spends the same amount of time in all the pipeline stages. The cycle time of the pipeline is $\mathbf{4\ ns}$. How much time(in ns) is required to execute the above program without using branch prediction?

Your Answer: | Correct Answer: 64 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Consider a system with cache access time 20 ns and main memory access time 140 ns . If 60% operations are read operations and hit ratio is 90%. What is the effective access time if write through updation technique is used?

A. 75.2 ns

B. 76.4 ns
C. 83.2 ns
D. 84.4 ns

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Copyright & Stuff